Basic Details

Department: Artificial Intelligence and Machine Learning
Program: Final Year-AIML
Academic Year: 2026-2027
Course Name: Bachelor of Technology
Project Code: AIML_P19
Project Type: Major
Planned Submission: 20 Aug 2026
Submitted: 26 Oct 2026

Guide Details

Name: Dr.Rajeshwar Rao K
Email: krajeshwarraoset@sanjivani.edu.in
Designation: Associate Professor
Area of Expertise: Machine learning , Data science, Software Engineering

Project Details

Description:
Description This project aims to develop Jurispectra, an AI-powered legal precedent analysis platform that assists legal researchers in identifying and analyzing historical court cases relevant to a new case. The system analyzes an uploaded judgment to understand its facts, legal issues, statutes, evidence, and reasoning, and retrieves relevant historical precedents using a fine tuned legal language model and hybrid retrieval techniques. It further performs multi-level case comparison to identify similarities and critical differences, and analyzes whether the important legal principles and decisive factors of a previous case can be meaningfully applied to the new case. By utilizing Legal Knowledge Graphs, Retrieval-Augmented Generation (RAG), GraphRAG, and Multi-Agent AI, the system provides evidence-grounded legal analysis, precedent relationships, and an explainable legal research report through an interactive dashboard.

Background & Need:
1. Legal researchers often need to examine a large number of previous judgments to identify relevant precedents for a new legal case.
2. Existing legal search systems mainly focus on retrieving similar or relevant cases, requiring further manual analysis to determine their actual applicability.
3. Cases that appear similar may contain important differences in facts, evidence, legal provisions, court authority, or circumstances that can affect the applicability of a precedent.
4. Advances in Legal NLP, Large Language Models, Knowledge Graphs, RAG, and Multi-Agent AI enable the development of intelligent systems that can support deeper legal precedent analysis.
5. Platforms such as indiakanoon.org show cases that look similar but do not show which specific fact makes a precedent succeed or fail. Juri spectra's What-If Fact Simulator lets the researcher change a fact (for example, "demand notice not sent within 30 days") and see how the precedent's applicability changes, so alternative scenarios can be tested before filing or arguing a case.

Problem Statement:
1. Problem Statement Most existing legal search and AI-based systems focus on retrieving similar or relevant court cases, but provide limited support for understanding whether a historical precedent is actually applicable to a new case. Legal researchers often need to manually compare facts, legal issues, evidence, statutes, court authority, and previous judgments, making legal research time-consuming and complex. There is a need for an intelligent Legal Precedent Analysis Platform that can retrieve relevant historical cases, perform detailed case comparison, identify critical differences, and analyze whether the legal principles and decisive factors of previous cases can be meaningfully applied to a new case through a unified AI-based system.

Objectives:
1. Develop an AI-based legal precedent analysis platform that analyzes new court cases and retrieves relevant historical precedents using legal language models and intelligent retrieval techniques.
2. Perform multi-level case comparison to identify similarities, critical factual and legal differences, and important factors between a new case and historical precedents
3. Develop precedent transferability analysis to evaluate whether the legal principles and decisive factors of a historical case can be meaningfully applied to a new case.
4. Provide evidence-grounded legal research insights using Knowledge Graphs, RAG, GraphRAG, and Multi Agent AI through an interactive legal analysis dashboard
5. Develop a What-If Fact Simulator that lets the user modify or remove individual case facts and instantly shows how the precedent's applicability score and verdict change, identifying which facts are decisive.
6. Develop a Law-Change Impact Analysis feature to assess whether historical legal precedents remain applicable under amended or replaced laws, including the transition from IPC to BNS.

Scope:
The proposed system focuses on developing a prototype of Jurispectra, an Intelligent Legal Precedent Analysis Platform that provides AI-assisted legal case retrieval, comparison, and precedent analysis within a unified environment. The platform analyzes new court cases by understanding facts, legal issues, statutes, evidence, and reasoning, and retrieves relevant historical precedents using legal language models and intelligent retrieval techniques. It performs multi-level case comparison, critical difference detection, precedent transferability analysis, and analysis of court authority, legal evolution, and conflicting precedents. The system further utilizes Knowledge Graphs, RAG, GraphRAG, and Multi-Agent AI to provide evidence-grounded legal research insights through an interactive dashboard. and a What-If Fact Simulator for testing the effect of changed facts on precedent applicability. The system will analyze changes between historical and current legal provisions, such as IPC and BNS, to identify whether changes in legal conditions may affect the applicability of earlier judicial precedents.

Limitations / Exclusions:
1. The project is limited to a prototype implementation and evaluation using publicly available and authorized Indian legal datasets and court judgments. 2. Complete coverage of all Indian courts, legal domains, jurisdictions, and large-scale professional legal deployment is beyond the scope of this project. 3. The system provides AI-assisted legal research and precedent analysis but does not replace legal professionals, and its accuracy depends on the quality of available legal data and the underlying AI models.

Methodology

Research Gap / Novelty:
Existing work demonstrates individual building blocks: knowledge-graph multi-agent verification (LegalGraphRAG), RAG with retrieved precedents for Indian law (NyayaRAG, AQgR), and Neo4j-based legal graphs (Domain-Partitioned Hybrid RAG). None of them takes a new judgment and decides, factor by factor, whether a specific precedent can be applied to it. Jurispectra does not claim novelty in using KG, RAG, or multi-agent AI. Its contribution is a structured, auditable Precedent Transferability Analysis for Indian law, which combines (i) decisive-factor matching between the precedent's ratio and the new case, (ii) distinguishing-difference detection, (iii) court-hierarchy authority under Indian law, and (iv) legal-evolution status (followed, distinguished, overruled, or legislatively superseded) derived from the citation graph. It outputs a graded verdict with evidence for each factor instead of a free-text explanation.

factors of a precedent, checks each one against the new case, and gives a graded verdict with evidence instead of a free-text explanation. 2. Authority and legal-evolution modeling. It uses court hierarchy and citation treatment (followed, distinguished, overruled, legislatively superseded) from a Legal Knowledge Graph to weaken precedents that are no longer good law. 3. What-If Fact Simulator. The user can change or remove a fact and instantly see how the applicability score changes, which shows exactly which fact makes a precedent work or fail. Platforms like indiakanoon.org do not offer this. 4. Evidence-verified output. Every finding is checked against the source judgment by a verification agent.

Proposed Solution:

We propose Jurispectra, an AI-powered Legal Precedent Analysis Platform for Indian courts. A researcher uploads a judgment or case details. The system then: Unlike indiakanoon.org, which returns documents for the user to analyze manually, Jurispectra tells the researcher whether a precedent applies, why, and which facts it depends on. It is a research aid and does not replace legal professionals.

Proposed Methodology Steps:

1. Collect and preprocess Indian legal case data, including historical judgments, case information, legal issues, statutes, and citation relationships.
2. Fine-tune a pretrained legal language model and develop hybrid retrieval using semantic, keyword, and citation-based information to identify relevant historical precedents.
3. Analyze and compare cases across facts, legal issues, statutes, evidence, circumstances, and reasoning to identify similarities and critical differences.
4. Law-Change Impact Analysis: Map relevant IPC provisions to corresponding BNS provisions, compare their legal ingredients and conditions, and assess whether the reasoning of historical judgments remains applicable under the new law.
5. Develop a What-If Fact Simulator that treats each decisive factor as an adjustable input (Satisfied / Absent / Contradicted). When the user changes a factor, the transferability score is recomputed and the system explains the change, naming the factor and the reason from the precedent's ratio.
6. Generate evidence-grounded legal research reports and evaluate the system using retrieval performance, comparison accuracy, transferability analysis, and baseline model comparisons.
7. Training and evaluation data will come from publicly available and authorized Indian legal sources: the open Supreme Court of India judgments dataset, High Court judgments from public eCourts portals, and existing research datasets such as ILDC, IL-PCSR and the NyayaRAG dataset. A pretrained Indian legal model (e.g., InLegalBERT) will be fine-tuned on this data.

Detailed Methodology:

Jurispectra accepts a court judgment or case details as input and extracts its facts, legal issues, statutes, evidence, and reasoning using Legal NLP. It preprocesses the judgment and uses a legal language model with hybrid semantic, keyword, and citation-based retrieval to identify and rank relevant historical precedents. The retrieved cases are compared to identify similarities, critical differences, and decisive legal factors. A Law-Change Impact Analysis module compares old and new legal provisions, including IPC-to-BNS transitions, to assess whether changes in legal conditions affect the reasoning of historical judgments. The Precedent Transferability Analysis evaluates each decisive factor, considering court authority and legal evolution, including whether a precedent has been followed, distinguished, overruled, or superseded, using a Neo4j Legal Knowledge Graph. A What-If Fact Simulator recalculates the applicability score when users modify case facts. Specialized AI agents, supported by RAG and Graph RAG, verify findings against original judgments and legal provisions, classify precedents as potentially applicable, requiring caution, or needing further legal review, and generate an evidence-grounded legal research report through the Juri spectra dashboard.

Literature Review (5 entries)

#	Author / Source	Year	Key Finding / Method	Gap Identified
1	Zerui Chen et al.	2026	Hierarchical legal knowledge graph with a three-agent pipeline (Researcher, Auditor, Adjudicator) for verified legal reasoning.	Not India-specific. It verifies answers to legal questions and does not compare a new case against precedents, model court hierarchy, or score transferability.
2	Shubham Kumar Nigam et al.	2025	Gives an LLM case facts, statutes and top-k retrieved precedents to predict Indian judgments.	The goal is outcome prediction. It uses no knowledge graph and no factor-level check of whether a precedent's reasoning applies.
3	Vishnuprabha V	2025	Question-guided RAG with structured summaries to show how a precedent applies to given facts.	The applicability is an LLM-generated explanation. It has no explicit score, no authority or overruling modeling, and no graph.
4	Rakshita Goel	2026	Neo4j graph of cases, statutes, IPC sections, judges and citations, plus three specialised RAG pipelines. It reports a 70% pass rate against 37.5% for RAG-only on a 40-question benchmark.	It is built for question answering. It does not take an uploaded judgment, compare cases, or decide applicability.
5	Shounak Paul et al.	2025	Indian prior-case and statute retrieval using lexical, semantic and graph methods.	Retrieval only. It does not say whether a retrieved case can be applied.

Proposed Assessment Criteria (8 entries)

			baseline	
Case Ranking	Qualitative	MRR / nDCG for ranking relevant precedents	Improvement over baseline	Ranking evaluation report
Case Comparison	Quantitative	Accuracy in identifying similarities and critical differences	≥ 80%	Case comparison evaluation
Precedent Transferability	Quantitative	Accuracy / F1-score in assessing precedent applicability	≥ 80%	Transferability evaluation report
Evidence Verification	Qualitative	Percentage of AI-generated findings supported by legal sources	≥ 90%	Verification report
Jurispectra Dashboard	Quantitative	Clarity and usefulness of case analysis and generated reports	≥ 4/5	User feedback survey
System Response Time	Quantitative	Time to generate case retrieval and analysis results	≤ 10 seconds	System logs
What-If Simulation	Quantitative	% of fact changes where the score moves in the direction a legal annotator expects	≥ 80%	What-if evaluation report

Resource Requirements (1 items)

Type	Details	Amount / Est. Value	Purpose	Contact Person
Hardware	Laptop/Desktop (16GB RAM recommended), Internet Connection	Existing	Model development, testing, and deployment	Project Team

Ethical, Safety & Permission Requirements

- ✓ **AI-generated content or AI-assisted writing**
- × Animal study
- × Confidential data
- × External organization data
- × Hazardous material
- ✓ **Human participants / survey** — Your plan includes a user feedback survey (dashboard ≥ 4/5) and law students or advocates labelling the gold set. Keep it to feedback and annotation, with no personal data collected.
- × Patient / medical data

Expected Deliverables

#	Deliverable	Details
1	Project report	Final project documentation including methodology, implementation, evaluation, and results.
2	Prototype / model / software / design	AI-powered legal precedent analysis platform with case retrieval, comparison, transferability analysis, Knowledge Graph, RAG, and Multi-Agent AI.
3	Dataset / survey / experimental result	Interactive dashboard for case upload, relevant precedent retrieval, case comparison, transferability analysis, and evidence-grounded legal reports.
4	Research paper / conference paper	Draft research paper describing the proposed methodology, research gap, novelty, and experimental evaluation.
5	Project Presentation	Final PPT and live demonstration of the Jurispectra system for project evaluation
6	Monitoring Dashboard	Interactive dashboard for case upload, relevant precedent retrieval, case comparison, transferability analysis, and evidence-grounded legal reports

Publication / Patent Plans

Output Type	Tentative Title	Venue / Journal	Expected Timeline	Details
Conference Paper	Jurispectra: Intelligent Legal Precedent Analysis	IEEE Regional Conference	IEEE International Conference on Sustainable AI, Networks, Knowledge and Learning Paradigms (SANKALP 2027), ABV-IIITM Gwalior, Madhya Pradesh, India, 2-4 April 2027	-

Work Plan & Timeline

MILESTONE	Phase 1	Literature survey, study of existing Legal AI systems, legal case retrieval, Legal NLP, and precedent analysis approaches. Finalize the problem definition, research gap, project requirements, and technology selection.	1 weeks	13 Oct 2026	Problem Definition Professionalism & Ethics Project Planning & Execution
MILESTONE	Phase 2	Legal dataset preparation and preprocessing pipeline. Collect publicly available or authorized Indian court judgments and extract facts, legal issues, statutes, evidence, reasoning, and citation information.	1 weeks	26 Oct 2026	Documentation Quality Problem Solving
MILESTONE	Phase 3	Legal AI model architecture design, selection of a pretrained legal language model, case representation, feature extraction, and initial development of the hybrid precedent retrieval system.	1 weeks	02 Nov 2026	Model Selection Problem Solving
MILESTONE	Phase 4	Initial legal model development and testing, semantic and keyword-based precedent retrieval, case ranking, and baseline performance evaluation using relevant retrieval metrics.	1 week	09 Nov 2026	Model Training Project Planning & Execution Teamwork & Collaboration
SEMESTER BREAK	**End of Semester 1 — Marks Evaluation Point**	Semester evaluation based on completed milestones before the break. Work after the break continues in the next semester cycle. Marks for this semester are calculated based on milestones completed before this break. **Marks for this semester are calculated on milestones completed before this break.**	Break period	18 Dec 2026	-
MILESTONE	Phase 5	Develop multi-level case comparison and precedent transferability analysis to identify similarities, critical factual and legal differences, and assess whether legal principles and decisive factors from historical cases can be applied to a new case.	2 weeks	19 Jan 2027	Model Selection Model Training Problem Solving Professionalism & Ethics
MILESTONE	Phase 6	Develop the Legal Knowledge Graph representing relationships among cases, laws, courts, and citations. Integrate RAG, GraphRAG, and Multi-Agent AI for contextual legal reasoning and evidence verification.	2 weeks	02 Feb 2027	Documentation Quality Presentation Skills Teamwork & Collaboration
MILESTONE	Phase 7	System integration and dashboard development, including case upload, precedent retrieval, case comparison, transferability analysis, API integration, interactive legal research reports, system testing, and deployment.	1 weeks	09 Feb 2027	Deployment & Inference Presentation Skills Teamwork & Collaboration
MILESTONE	Phase 8	Error analysis, false-positive and false-negative checking, evaluation of retrieval performance, case comparison accuracy, transferability accuracy, evidence verification, response time, and overall system improvement.	2 weeks	23 Feb 2027	Documentation Quality Overall Project Quality Problem Solving
MILESTONE	Phase 9	Final report preparation, code finalization, technical documentation, results summary, research paper preparation, project presentation, and final demonstration of the Jurispectra platform.	1 weeks	30 Mar 2027	Communication Skills Documentation Quality Presentation Skills

Student Details (4 students)

Student 1

Name: **Rucha Dange**
Email: rucha.dange_24uml@sanjivani.edu.in
Responsibility: Project planning, legal-AI research, dataset preparation, model fine-tuning, system architecture, and overall project integration.

PRN / Roll No.: 2124UMLF2053
Mobile: 9527487575
Expected Contribution: (25%)Project planning, legal-AI research, dataset preparation, model fine-tuning, system architecture, and overall project integration.

Student 2

Name: **Vedika Taji**
Email: sachin.vedika_24uml@sanjivani.edu.in
Responsibility: Frontend development, Jurispectra Dashboard, UI/UX design, and API integration.

PRN / Roll No.: 2124UMLM2073
Mobile: 8600749570
Expected Contribution: (25%)Frontend development, Jurispectra Dashboard, UI/UX design, and API integration.

Student 3

Name: **Sakshi Abak**

PRN / Roll No.: 2124UMLF2019

Student 4

Name: **Swaroop Gade**
PRN / Roll No.: 2124UMLM2032
Email: swaroop.gade_24uml@sanjivani.edu.in
Mobile: 8767728185
Responsibility: Knowledge Graph, RAG or GraphRAG, AI agent integration, testing, deployment, documentation.
Expected Contribution: (25%)Knowledge Graph, RAG or GraphRAG, AI agent integration, testing, deployment, documentation.

Workflow History

Under Research Review → Approved 23 hours ago
by Arvind Mukundan
Jurispectra addresses a meaningful problem and has potential as a strong final-year project. However, the proposed combination of RAG, knowledge graphs, agents, fine-tuning, legal-change analysis, and simulation is too broad for the available timeline. Please select one Indian legal domain and define a transparent, reproducible precedent-transferability method. Specify the exact dataset, expert annotation procedure, decision rules, and baseline systems. Evaluate retrieval, factor identification, applicability classification, and evidence correctness separately, including failure cases. The paper should demonstrate measurable improvement over existing legal-AI approaches. Approval is recommended subject to these methodological revisions and a realistic implementation plan and legal validation.

Submitted (Pending Audit) → Under Research Review 2 days ago
by Nilesh Gaikwad
Conditional approval on the basis of outcome

Pending Guide Review → Submitted (Pending Audit) 4 days ago
by Dr.Rajeshwar Rao K
changes done, please approve the proposal

Revision Requested → Pending Guide Review 4 days ago
by Rucha Dange
Resubmitted after revision. Sections updated: Step 9, Publication & Plan, Students, Project & Problem, Research Proposal, Resources & Ethics, Outcomes.

Submitted (Pending Audit) → Revision Requested 6 days ago
by Nilesh Gaikwad
Please refer the milestone template. Due to restriction of milestone issues, cannot forward if milestone > 2 weeks

Pending Guide Review → Submitted (Pending Audit) 1 week ago
by Dr.Rajeshwar Rao K
as per the suggestions from auditor , updated milestones and Project to be completed by End of March 2027 .

Revision Requested → Pending Guide Review 2 weeks ago
by Rucha Dange
Resubmitted after revision. Sections updated: Step 9, Basic Details, Guide Info, Project & Problem, Research Proposal, Resources & Ethics, Outcomes, Publication & Plan, Students.

Submitted (Pending Audit) → Revision Requested 3 weeks ago
by Nilesh Gaikwad
List of papers available:
LegalGraphRAG: Multi-Agent Graph Retrieval-Augmented Generation for Reliable Legal Reasoning. Authors: Zerui Chen et al. (Xiamen University group). Venue: ACL 2026 (Long Papers), also arXiv:2605.28120. Hierarchical legal knowledge graph plus a three-agent pipeline (Researcher/Auditor/Adjudicator); code at github.com/XMUDeepLIT/LegalGraphRAG. Conference paper, no SCImago quartile applies.
NyayaRAG: Realistic Legal Judgment Prediction with RAG under the Indian Common Law System. Authors: Shubham Kumar Nigam et al. Venue: IJCNLP/ACL Anthology 2025, arXiv:2508.00709. Feeds an LLM case facts + statutes + top-k retrieved precedents; code at github.com/ShubhamKumarNigam/NyayaRAG. Conference paper, quartile not applicable.
Augmented Question-guided Retrieval (AQgR) of Indian Case Law with LLM, RAG, and Structured Summaries. arXiv:2508.04710. Aims to show how a precedent applies to given facts (core of 'transferability'), not just ranking similarity. Preprint, quartile not verified.
Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India. arXiv:2602.23371. Builds a Neo4j legal knowledge graph (cases, statutes, IPC sections, judges, citations) plus three specialized RAG pipelines; reports 70% pass rate vs 37.5% for RAG-only baseline on a 40-question benchmark. Preprint, quartile not verified.
List of Github / Kaggle Dataset links:
JusticeRAG (github.com/sassysohom48/JusticeRAG) — retrieval-augmented legal precedent discovery for Indian case law; FastAPI, Next.js, Qdrant, BM25+embeddings+LLM; MIT license, live demo link in README. Does not clearly show a knowledge-graph component.
NyayaRAG (github.com/ShubhamKumarNigam/NyayaRAG) — dataset and RAG pipeline code for Indian legal judgment prediction, linked from the paper above.
LegalGraphRAG (github.com/XMUDeepLIT/LegalGraphRAG) — code/datasets for the hierarchical-graph + multi-agent legal reasoning system (not India-specific, but directly relevant to the KG+multi-agent architecture Jurispectra proposes).
Literature gap and issues:
The claimed gap is only partially open. LegalGraphRAG already combines a hierarchical legal knowledge graph with a multi-agent (researcher/auditor/adjudicator) verification pipeline, and NyayaRAG/AQgR already use RAG specifically to show how a precedent applies to new facts for Indian law — essentially what 'precedent transferability' means. No single system combines all of KG + RAG + multi-agent + an explicit Indian-specific transferability score + court-authority modeling, so the narrow 'unified framework' claim still holds, but each individual building block has already been demonstrated elsewhere. The student needs to be precise about what is genuinely new versus re-assembling existing, published components.
List of Questions:
1. LegalGraphRAG already uses a hierarchical knowledge graph and a researcher/auditor/adjudicator multi-agent pipeline for verified legal reasoning — walk me through exactly what your multi-agent pipeline adds beyond what their auditor agent already checks.

underlying data come from?

4. Since no public Indian benchmark exists for 'is this precedent transferable', how will you validate that your system's judgment matches what a real lawyer would conclude?
5. Walk me through one end-to-end example — an uploaded judgment, the precedents it retrieves, and the actual decision logic your system uses to decide transferability, not just the high-level architecture labels.

Based on the above data, please revise the literature review and methodology accordingly.

Revision Requested → Submitted (Pending Audit) 3 weeks ago
by Dr.Rajeshwar Rao K
students are suggested to do the modifications

Submitted (Pending Audit) → Revision Requested 1 month ago
by Nilesh Gaikwad
Need more detailed analysis and also mention advantages of your application over indiakanoon as they are also using AI.

Pending Guide Review → Submitted (Pending Audit) 1 month ago
by Dr.Rajeshwar Rao K
please check sir

Revision Requested → Pending Guide Review 1 month ago
by Rucha Dange
Proposal submitted.

Submitted (Pending Audit) → Revision Requested 2 months ago
by Nilesh Gaikwad
Please correct milestones. Milestone duration should be min 1 and max 2 weeks.

Pending Guide Review → Submitted (Pending Audit) 2 months ago
by Dr.Rajeshwar Rao K
ok , forwarded to Auditor

Draft → Pending Guide Review 2 months ago
by Rucha Dange
Proposal submitted.

Draft (initialized) 2 months ago
Workflow initialized.

Study Flowchart

